

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
1	(a)	(i)		Same distance between lamp and LDR (1) Same {ambient / background / room } lighting (1)	2			2		2
		(ii)		Enclose lamp and LDR in a container e.g. tube / carry out experiment in a darkened room (1) So no ambient light affects LDR (1)			2	2		2
	(b)	(i)		Scales: y-axis: 2 k Ω per 2 cm (1) The 6 points between $P = 4$ and 24 W plotted correctly to within $\pm <$ small square division (2) 5 correct plots (1) Smooth curve of best fit between $P = 4$ and 24 W with $\pm <$ small square division tolerance; may miss one point (1) [No hairy lines] Non-linear scale $\rightarrow$ total mark 0.	1	2 1		4	4	4
		(ii)		Read from graph 2600 Ω / 2.6 k Ω		1		1	1	1
		(iii)		From 12 [W] to 24 [W] the resistance goes from 2.2 [k Ω] to 1.1 [k Ω], or 8 to 16, from 3 to 1.5 (1) Other doubling e.g. 2 [W] to 4 [W] resistance goes from 19.5 [k Ω] to 10.3 [k Ω] or 4 to 8, from 10.3 to 3 (1) Therefore sometimes true, sometimes not or w.t.t.e (1) NB 3rd mark only given following arguments for first and second mark presented (whether marks awarded or not)			3	3	1	3

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	(c)			Substitution: $0.35 = \frac{2.8}{R}$ (1) Manipulation: $R = \frac{2.8}{0.35}$ (1) Resistance = 8 000 [Ω] or 8 k[Ω] (1) $R = 8 \times 10^n$ ($n \neq 3$) $\rightarrow$ 2 marks $2.8 \times 0.35 = 0.98 \rightarrow$ 0 marks [unless first mark awarded]	1	1 1		3	3	
				Question 1 total	4	6	5	15	9	12